

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI
B.E. in Chemical Engineering
Scheme of Teaching and Examinations 2022
 Outcome Based Education (OBE) and Choice Based Credit System (CBCS)
 (Effective from the academic year 2023-24)

V SEMESTER

Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing	SDA	Duration in hours	CIE Marks	SEE Marks	Total Marks	
					L	T	P	S					
1	HSMS	BCH501	Industrial Process Management	CH	3	0	0		03	50	50	100	3
2	IPCC	BCH502	Chemical Reaction Engineering	CH	3	0	2		03	50	50	100	4
3	PCC	BCH503	Mass Transfer Operations-I	CH	4	0	0		03	50	50	100	4
4	PCCL	BCHL504	Mass Transfer Operations Lab-1	CH	0	0	2		03	50	50	100	1
5	PEC	BCH515x	Professional Elective Course	CH	3	0	0		03	50	50	100	3
6	PROJ	BCH586	Mini Project	CH	0	0	4		03	100		100	2
7	AEC	BRMK557	Research Methodology and IPR	Any Department	2	2	0		02	50	50	100	3
8	MC	BESK508	Environmental Studies	Any Department	2	0	0		02	50	50	100	2
9	MC	BNSK559	National Service Scheme (NSS)	NSS coordinator	0	0	2			100		100	0
		BPEK559	Physical Education (PE) (Sports and Athletics)	Physical Education Director									
		BYOK559	Yoga	Yoga Teacher									
Total									500	300	800	22	
Professional Elective Course													
BCH515A		Polymer Science &Engineering			BCH515C		Material Science and Engineering						
BCH515B		Industrial Pollution Control			BCH515D		Chemical Equipment Design						

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					Theory Lecture	Tutorial	Practical/ Drawing	SDA	Duration in hours	CIE Marks	SEE Marks	Total Marks	
1	IPCC	BCH601	Process Equipment Design and Drawing	CH	3	0	2		03	50	50	100	4
2	PCC	BCH602	Mass Transfer Operations-II	CH	4	0	0		03	50	50	100	4
3	PEC	BCH613x	Professional Elective Course	CH	3	0	0		03	50	50	100	3
4	OEC	BCH654x	Open Elective Course		3	0	0		03	50	50	100	3
5	PROJ	BCH685	Project Phase I	CH	0	0	4		03	100	--	100	2
6	PCCL	BCHL606	Mass Transfer Operations lab-2	CH	0	0	2		03	50	50	100	1
7	AEC/SDC	BCH657x	Ability Enhancement Course/Skill Development Course V	CH	If the course is offered as a Theory				01	50	50	100	1
					1	0	0						
					If course is offered as a practical								
					0	0	2						
8	MC	BNSK658	National Service Scheme (NSS)	NSS coordinator	0	0	2			100	---	100	0
		BPEK658	Physical Education (PE) (Sports and Athletics)	Physical Education Director									
		BYOK658	Yoga	Yoga Teacher									
9	IKS	BIKS609	Indian Knowledge System		1	0	0		01	100	0	100	0
Total									500	300	800	18	
Professional Elective Course													
BCH613A		Chemical Process Industries			BCH613C		Chemical Process Integration						
BCH613B		Applied Mathematics in Chemical Engineering			BCH613D		Petroleum Refinery Engineering						
Open Elective Course													
BCH654A		Nano Science and Technology			BCH654C		Solid Waste Management in Process Industries						
BCH654B		Principles of Downstream Techniques in Bio process			BCH654D		Material Science and Technology						
Ability Enhancement Course / Skill Enhancement Course-V													
BCH657A		Technical Report writing and Presentation skills			BCH657C		Problem Based learning						
BCH657B		Introduction to start-ups			BCH657D		Project Management						

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VII SEMESTER (Swappable VII and VIII SEMESTER) SCHEME A													
Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question and Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing	SDA	Duration in hours	CIE Marks	SEE Marks	Total Marks	
1	IPCC	BCH701	Process Modeling and Simulation	CH	3	0	2		03	50	50	100	4
2	IPCC	BCH702	Process control and Industrial IOT	CH	3	0	2		03	50	50	100	4
3	PCC	BCH703	Process Engineering Economics and Management	CH	4	0	0		03	50	50	100	4
4	PEC	BCH714x	Professional Elective Course / (Online)	CH	3	0	0		03	50	50	100	3
5	OEC	BCH755x	Open Elective Course /(online)		3	0	0		01	50	50	100	3
6	PROJ	BCH786	Major Project Phase-II	CH	0	0	12		03	100	100	200	6
										400	300	700	24
Professional Elective Course													
BCH714A		Transport Phenomena			BCH714C		Pharmaceutical Technology						
BCH714B		Industrial Safety and Risk Management			BCH714D		Biochemical Engineering						
Open Elective Course													
BCH755A		Disaster Management			BCH755C		Air Pollution Control Techniques						
BCH755B		Sustainable Technology			BCH755D		Protein Chemistry						
PCC: Professional Core Course, PCCL: Professional Core Course laboratory, PEC: Professional Elective Course, OEC: Open Elective Course PR: Project Work, L: Lecture, T: Tutorial, P: Practical S= SDA: Skill Development Activity, CIE: Continuous Internal Evaluation, SEE: Semester End Evaluation. TD- Teaching Department, PSB: Paper Setting department, OEC: Open Elective Course, PEC: Professional Elective Course. PROJ: Project work													
Note: VII and VIII semesters of IV years of the program													
(1) Institutions can swap the VII and VIII Semester Schemes of Teaching and Examinations to accommodate research internships/ industry internships after the VI semester.													
(2) Credits earned for the courses of VII and VIII Semester Scheme of Teaching and Examinations shall be counted against the corresponding semesters whether the VII or VIII semesters is completed during the beginning of the IV year or the later part of IV years of the program.													

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VIII SEMESTER (Swappable VII and VIII SEMESTER)

Sl. No	Course and Course Code		Course Title	Teaching Department (TD) and Question Paper Setting Board (PSB)	Teaching Hours /Week				Examination				Credits
					Theory Lecture	Tutorial	Practical/ Drawing	SDA	Duration in hours	CIE Marks	SEE Marks	Total Marks	
1	PEC	BCH801x	Professional Elective (Online Courses)		3	0	0		03	50	50	100	3
2	OEC	BCH802x	Open Elective (Online Courses)		3	0	0		01	50	50	100	3
3	INT	BCH803	Internship (Industry/Research) (14 - 20 weeks)		0	0	12		03	100	100	200	10
										200	200	400	16

Professional Elective Course (Online courses)

BCH801A	(Online Courses)	BCH801C	(Online Courses)
BCH801B	(Online Courses)	BCH801D	(Online Courses)

Open Elective Courses (Online Courses)

BCH802A	(Online Courses)	BCH802C	(Online Courses)
BCH802B	(Online Courses)	BCH802D	(Online Courses)

L: Lecture, **T:** Tutorial, **P:** Practical **S= SDA:** Skill Development Activity, **CIE:** Continuous Internal Evaluation, **SEE:** Semester End Evaluation. **TD-** Teaching Department, **PSB:** Paper Setting department, **OEC:** Open Elective Course, **PEC:** Professional Elective Course. **PROJ:** Project work, **INT:** Industry Internship / Research Internship / Rural Internship

Note: VII and VIII semesters of IV years of the program

Swapping Facility

- Institutions can swap VII and VIII Semester Scheme of Teaching and Examinations to accommodate **research internships/ industry internships/Rural Internship** after the VI semester.
- Credits earned for the courses of VII and VIII Semester Scheme of Teaching and Examinations shall be counted against the corresponding semesters whether VII or VIII semester is completed during the beginning of IV year or later part of IV year of the program.

Elucidation:

At the beginning of IV years of the program i.e., after VI semester, VII semester classwork and VIII semester **Research Internship /Industrial Internship / Rural Internship** shall be permitted to be operated simultaneously by the University so that students have ample opportunity for an internship. In other words, a good percentage of the class shall attend VII semester classwork and a similar percentage of others shall attend to Research Internship or Industrial Internship or Rural Internship.

Research/Industrial /Rural Internship shall be carried out at an Industry, NGO, MSME, Innovation center, Incubation centre, Start-up, centre of Excellence (CoE), Study Centre established in the parent institute and /or at reputed research organizations/institutes.

The mandatory Research internship /Industry internship / Rural Internships for 14 to 20 weeks. The internship shall be considered as a head of passing and shall be considered for the award of a degree. Those, who do not take up/complete the internship shall be declared to fail and shall have to complete it during the subsequent University examination after satisfying the internship requirements.

Research internship: A research internship is intended to offer the flavor of current research going on in the research field. It helps students get familiarized with the field and imparts the skill required for carrying out research.

Industry internship: Is an extended period of work experience undertaken by students to supplement their degree for professional development. It also helps them learn to overcome unexpected obstacles and successfully navigate organizations, perspectives, and cultures. Dealing with contingencies helps students recognize, appreciate, and adapt to organizational realities by tempering their knowledge with practical constraints.

Rural Internship: Rural development internship is an initiative of Unnat Bharat Abhiyan Cell, RGIT in association with AICTE to involve students of all departments studying in different academic years for exploring various opportunities in techno-social fields, to connect and work with Rural India for their upliftment.

The faculty coordinator or mentor has to monitor the student's internship progress and interact with them to guide for the successful completion of the internship.

The students are permitted to carry out the internship anywhere in India or abroad. University shall not bear any expenses incurred in respect of the internship.

With the consent of the internal guide and Principal of the Institution, students shall be allowed to carry out the internship at their hometown (**within or outside the state or abroad**), provided favorable facilities are available for the internship and the student remains regularly in contact with the internal guide. **University shall not bear any cost involved in carrying out the internship by students.** However, students can receive any financial assistance extended by the organization.

Professional Elective /Open Elective Course:These are ONLINE courses suggested by the respective Board of Studies. Details of these courses shall be made available for students on the VTU web portal.